

Sami Ben Brahim

- Email: sami DOT benbrahim AT mail DOT concordia DOT ca
 - [LinkedIn](#)
 - [Github](#)
 - [Web](#)
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Education

PhD student: Information and Systems Engineering *Sep. 2023 – ongoing*
Concordia University Montreal, Canada
Supervisors: Nizar Bouguila and Manar Amayri

Master of Engineering: Information Processing and Complexity of the Living *Sep. 2020 – Nov. 2021*
ENIT Tunis, Tunisia
Thesis: Natural Gas Peak Demand Forecasting in Tunisia

Note: In Tunisia, the Master of Engineering is a one-year program that can be completed in parallel with the final year of the Bachelor of Engineering. The Bachelor's and Master's theses are typically the same.

Bachelor of Engineering: Modeling for Industry and Service (MIndS) *Sep. 2018 – Sep. 2021*
ENIT Tunis, Tunisia

Research Interests

Time Series Forecasting, Electricity Load Forecasting, Representation Learning, Machine Learning.

Experience

Teaching Assistant *Jan. 2024 – ongoing*
Concordia University Montreal, Canada

- Teaching Assistant for the following courses: COMP 6721, COMP 432, COMP 425, COMP 333, COMP 233, INSE 6180, COEN 243, ENCS 282.
- Leading tutorials, Mentoring students, Grading exams/assignments/projects.

Lecturer *Nov. 2024*
Concordia University Montreal, Canada

- Delivered a lecture on Time Series Forecasting as part of INSE 6180 course.

Research Assistant & Teaching Assistant *Apr. 2022 – Mar. 2023*
Africa Business School Rabat, Morocco

- Literature Review on the Topic 'Lot-Sizing and Scheduling Under Uncertainty'
- Implementation of stochastic optimization models to estimate the arrival time of vessels.
- Preparation of Data Science Course Material, Grading assignments.

Research Intern *Dec. 2021 – Mar. 2022*
Chair for Energy System Economics (FCN-ESE), RWTH Aachen University Aachen, Germany

- Processing and analysis of geo-referenced data.
- Cluster analysis of residential buildings based on their longitude, latitude, and area-specific heat demand.

Publications

[Google Scholar](#)

- SpanPatchTST: A self-supervised learning framework for short- and mid-term forecasting of power transformer load and oil temperature.** Jun. 2026
 – Sami Ben Brahim, Manar Amayri, Nizar Bouguila, *Sustainable Energy, Grids and Networks*.
- One-day-ahead electricity load forecasting of non-residential buildings using a modified Transformer-BiLSTM adversarial domain adaptation forecaster.** Apr. 2025
 – Sami Ben Brahim, Manar Amayri, Nizar Bouguila, *International Journal of Dynamics and Control*.
- Long-term natural gas peak demand forecasting in Tunisia Using machine learning.** May. 2022
 – Sami Ben Brahim, Mohamed Slimane, *5th International Conference on Advanced Systems and Emergent Technologies (IC_ASET)*.

Relevant Courses & Certificates

- IFT 6135 Representation Learning (Instructor: Aaron Courville)
- COMP 6321 Machine Learning (Instructor: Yang Wang)
- INSE 6180 Security and Privacy Implications of Data Mining (Instructor: Nizar Bouguila)
- Deep Learning Specialization (Online, deeplearning.ai)
- Machine Learning (Online, Stanford University)

Programming Skills

- **Languages:** Python, MATLAB, C++, R, LaTeX.
- **Libraries:** Pytorch, TensorFlow, Keras, Scikit-Learn, Numpy, Pandas, OpenCV.
- **Technologies:** GitHub.

Languages

Arabic (Native), English (Fluent), French (Fluent).

Honours & Scholarships

Erasmus+ ICM Scholarship 2021
 Exchange semester at *Rouen University* — *Awarded but not undertaken due to COVID-19 travel restrictions*

Review Activities

- Reviewer at IEEE Transactions on Neural Networks and Learning Systems (Impact factor: 8.9)
- Reviewer at Engineering Applications of Artificial Intelligence (Impact factor: 8.0)

Other Activities & Volunteering

- Attended ISF 2026 (International Symposium on Forecasting)
- Volunteer at the IEEE PES GM 2026
- Volunteer at ISO Fall 2024 Office Welcome